

Web 3.0 Unveiled - Day 3

Mentored by FEC

What is GAS?

Watch this video:  What is Ethereum Gas? (Examples + Easy Explanation)

What is Mining?

Mining is the process that helps create new blocks of transactions to be added to the Ethereum blockchain network. It also helps keep the network secure from attacks.

Just like Bitcoin, Ethereum was using a Proof-of-Work (PoW) consensus mechanism, which is heavily reliant on mining. Ethereum miners used their time and computation power to process new incoming transactions and produce new blocks until Ethereum upgraded to Proof-of-Stake consensus mechanism (PoS).

FunFact: The [Ethereum Merge](#) which was executed on September 15, 2022, replaced Proof-of-Work with Proof-of-Stake. This significantly changed how new blocks are produced, [reducing energy consumption](#) by ~99.95%. The topic Proof-of-Stake will be discussed in a later course.

Why do miners exist?

Most people think that mining is simply a way of adding new transactions and creating new coins. But, we mentioned above, that mining also helps keep the network secure. Why is that?

In decentralized systems like Ethereum, we need to ensure that everyone in the world agrees on the ordering of transactions.

For example, if Alice sent Bob 1 ETH, which he further sent to Charlie, the order of the transactions NEEDS to be as follows

Alice --> Bob 1 ETH

Bob --> Charlie 1 ETH

Any other transaction order will not work, and it is important everyone agrees to this.

Miners are also responsible for validating cryptocurrency transactions on the blockchain network and adding them to the ledger. Importantly, this process prevents the [double-spending](#) of digital currency.

Why would anyone become a miner?

Just like physical currencies, when someone executes a transaction, the state of the blockchain must be updated to reflect the debit/credit of each account involved.

Digital platforms however are prone to attacks, and can be manipulated. Miners therefore not only propose new blocks and validate all transactions, they also **need to produce a certificate of legitimacy** that all transactions in their proposed block are valid.

Producing this certificate is a computationally hard and complex challenge. This is what all of Proof of Work is about. We will go more in depth about this in a later tutorial, but for now, know that producing this certificate is quite hard.

But, given a certificate of legitimacy, it is extremely easy for a third party to verify whether it is valid.

For the miners' hard work in producing these certificates and executing transactions, they are rewarded in the form of new coins. In Ethereum, when a miner successfully mines a block, they receive 2 ETH for their hard work. Ethereum started with a block reward of 5 ETH, but this was lowered to 3 ETH with the Byzantium upgrade in 2017, and then again down to 2 ETH with the Constantinople upgrade in 2019.

Since decentralized systems lack a central authority, **mining is crucial to the network's safety and validity**. Therefore, the mining reward acts as an incentive to participate in the transaction validation process.

Costs of Mining

- Costs of hardware necessary to build and maintain the hardware
- Electrical costs of powering the mining rig
- Potential cost of equipment to support the mining rig (coolers, ventilators, energy monitors, electrical wiring, solar farms, etc.)

How are Ethereum transactions mined?

1. A user signs a transaction request from their account
2. The transaction is **broadcasted to the Ethereum network** through a node
3. Upon hearing of the new transaction, each node in the network **adds the transaction to their local mempool** (memory pool). The **mempool is a list of transactions that the nodes have heard** about but have not yet been included as part of a block.
4. At some point, a miner groups a lot of transactions into a potential block, in a way that maximizes transaction fees they earn while staying under the block gas limits.
5. The miner **verifies the validity of each transaction**. This involves things like checking that no one is trying to transfer ether they don't own, the signature is invalid, the request is malformed, etc.
6. The miner then **executes the transaction on their local Ethereum Virtual Machine (EVM) copy**. It awards the transaction fees for each such request to its own account.
7. The miner begins creating the Proof-of-Work certificate of legitimacy for the entire block after all transactions in the block have been verified and executed on their local EVM.
8. Eventually, the miner **finishes producing the certificate of legitimacy for the block**, which includes the user's transaction. The miner then broadcasts the completed block to the world, which includes the certificate and the new state information for the blockchain.
9. Other nodes hear about the new block. They **verify the certificate** (which is easy to do), **execute all transactions in the block on their local copies of the EVM**, and verify that their EVM state matches the one proposed by the miner.
10. Once other nodes have verified everything proposed by the miner, they add this block to their blockchain node, and accept the new EVM state as **"the current state of the blockchain"**.
11. Each node removes the transactions included within the block from their local mempool.
12. This process repeats.

Different methods of mining?

In the technology's early days, when it was quite easy to become a miner, most miners used CPU-based mining running on their desktops and laptops. However, **CPU mining is quite slow and impractical today** because it can take months to earn even a tiny amount of mining rewards while constantly running your CPU at 100% of its capacity and paying for high electrical costs.

GPU-based mining is the most common approach these days. This approach can maximize the computation power by bringing together multiple GPUs under a single mining rig, putting them all to work in parallel, thereby increasing the overall capacity and throughput of the one rig.

Application-Specific Integrated Circuit (ASIC) mining is also quite popular. These are hardware chips designed specifically to mine Ethereum. You can purchase ASICs online for the most popular blockchain networks. However, they are expensive, and as more and more miners

join the network and mining becomes harder, the hardware becomes obsolete quickly. A brand new ASIC would usually only last 1-2 years max.

Another growing option is cloud mining, which allows individual miners to leverage the power of dedicated mining facilities without owning any hardware.